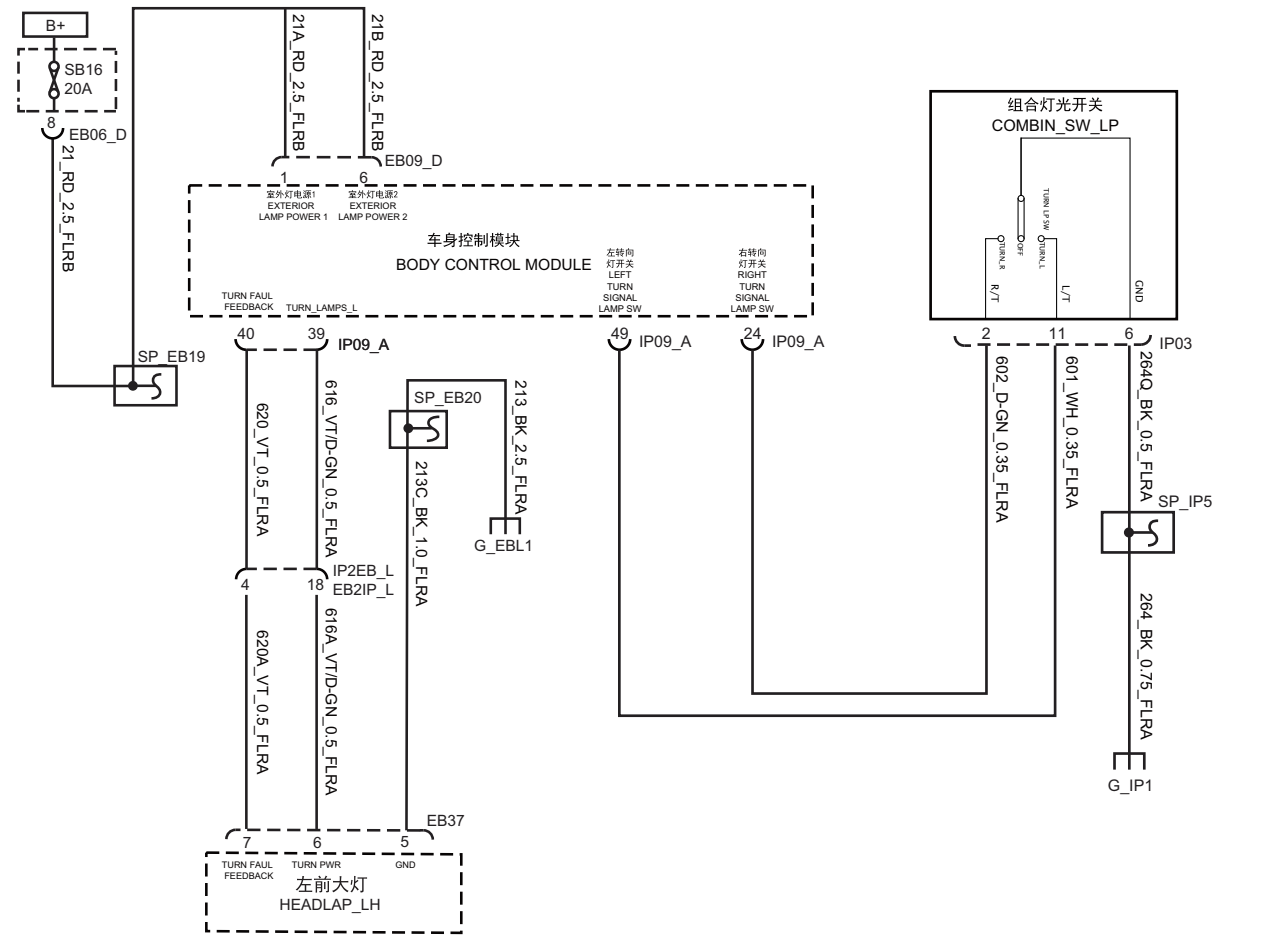
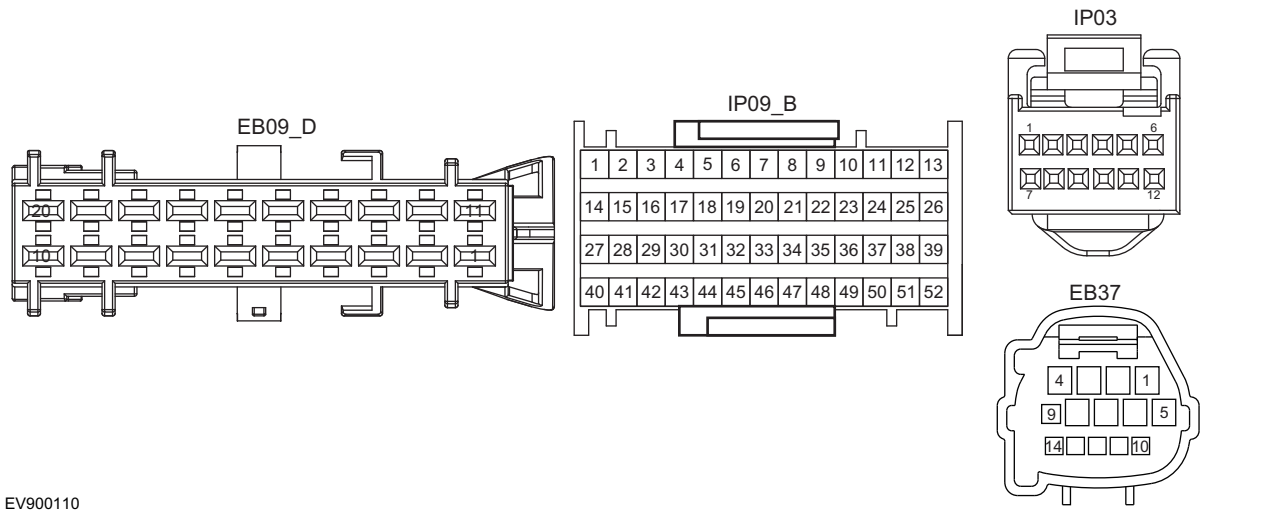
B1241-11、B1241-12、B1241-13

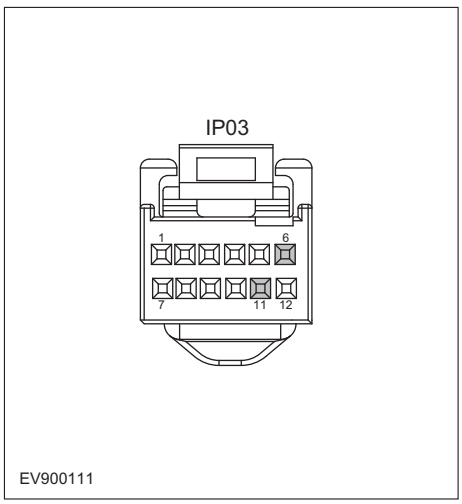
Fault diagnosis code
B1241-11：Left front turn light circuit short to ground
B1241-12：Left front turn light circuit short to battery
B1241-13：Left front turn light circuit open
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • BCM fault • Left front turn light circuit fault • Insurance fuse fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1241-11
BCM high beam light circuit open
B1241-12
Left front turn light output circuit short to battery
B1241-13
Left turn light output circuit open
To set the effect of a fault code condition
Left turn light function fails to effect
Description of circuit diagram

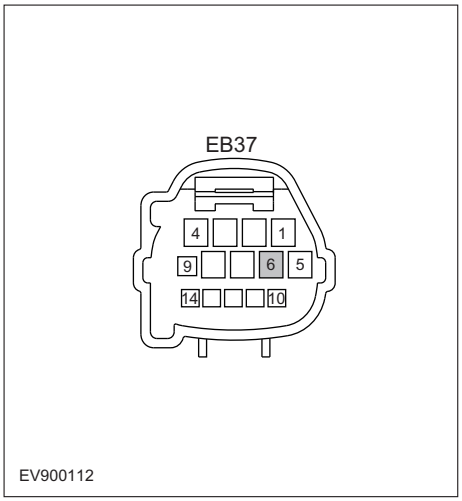
Circuit diagram

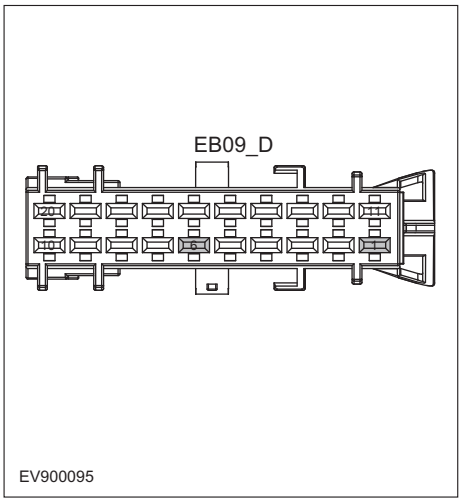


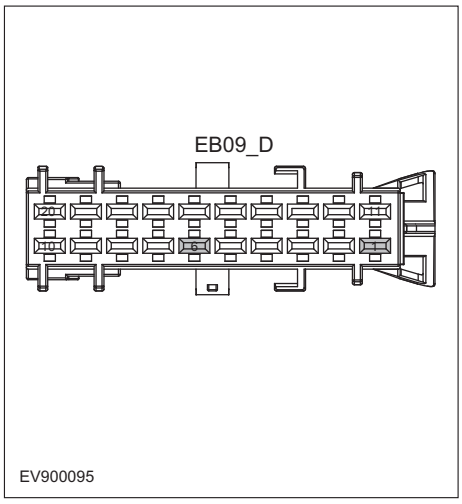
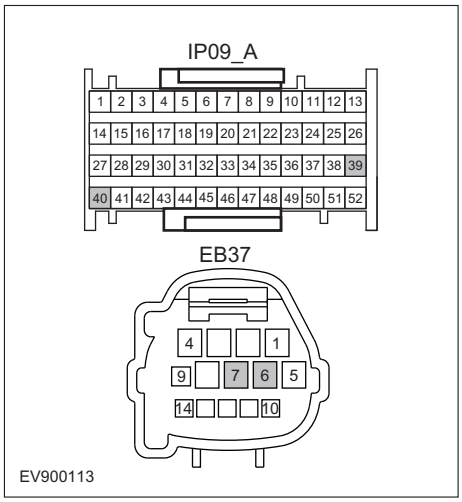


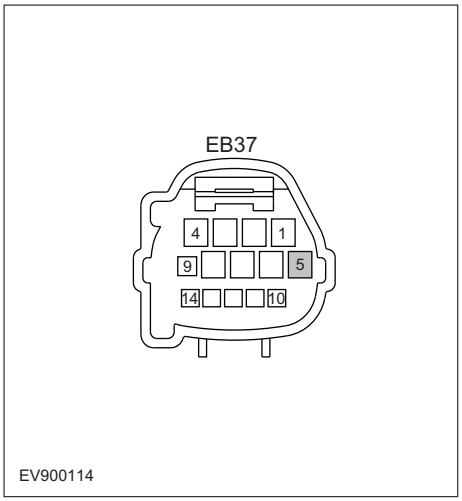
EV900110

Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse SB16 for damage. 2 Check battery capacity. 3 Check the harness plug of air conditioning control module for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Check the body control module with a diagnostic apparatus. 4 Is there any other fault code except for B1241-11/B1241-12/B1241-13?
		YES → Refer to:DTC Summary list(BCM).
NO ↓		
Step 3	Check the combination lights switch (turning light switch)(components).	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP03 of IPK module. 3 Place the combination light switch in the “left turn signal”position and measure the resistance between the combination light switch terminals(combination switch). Measuring circuit: resistance between terminal 11 on plug IP03 and terminal 6 on plug IP03. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
		NO → Replace combination lights.
YES ↓		

Step 4	Check the supply power of turn light when it is started or operating.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB37 of left front turn light. 3 Turn the ignition switch to ON. 4 Measure the voltage between the harness terminal of left front turn light and ground. Measuring circuit: voltage between terminal 6 on plug EB37 and ground. **Standard value:10~14V** 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 5	Check the power voltage of outdoor light of BCM
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug EB09_D of BCM. 3 Measure the voltage between the harness terminal of BCM and ground. Measuring circuit: voltage between terminal 1 on plug EB09_D and ground. Measuring circuit: voltage between terminal 6 on plug EB09_D and ground. **Standard value:10~14V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the power voltage of outdoor light when the BCM is started or operating
	1 Turn the ignition switch to ON. 2 Measure the voltage between the harness terminal of BCM and ground. Measuring circuit: voltage between terminal 1 on plug EB09_D and ground. Measuring circuit: voltage between terminal 6 on plug EB09_D and ground. **Standard value:10~14V** 3 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 7	Check the circuit between left front turn light and BCM (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP09_A of BCM. 3 Measure the resistance between the harness terminal of left front turn light and of BCM. Measuring circuit: resistance between terminal 6 on plug EB37 and terminal 39 on plug IP09_A. Measuring circuit: resistance between terminal 7 on plug EB37 and terminal 40 on plug IP09_A. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 8	Test whether the grounding circuit of turn light is open.
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness terminal of left front low beam light and ground. Measuring circuit: resistance between terminal 5 on plug EB37 and ground. **Standard value: < 1Ω** 3 Test whether the resistance is less than 1Ω?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 9	Test whether BCM is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BCM.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	